

A low-angle, upward-looking photograph of several tall skyscrapers. The buildings are dark and silhouetted against a bright, solid red sky. The perspective creates a sense of height and scale, with the buildings converging towards the top of the frame.

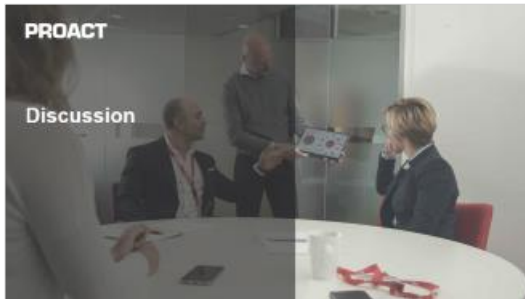
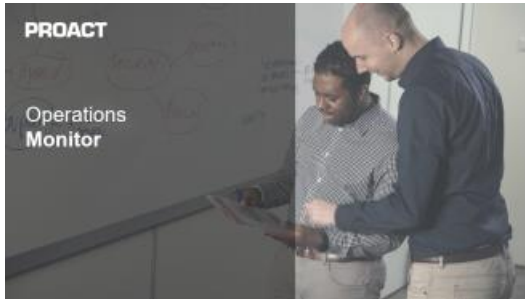
Foundation Workshop
OPERATIONS

PROACT

Azure Operations Session

PROACT

Agenda



Cloud Operations

Manage

Business Alignment

Criticality



Document the criticality and relative business value of each workload.

Impact



Establish clear performance expectations and business interruption time/value metrics.

Commitment



Document, track, and report on commitments to cost and performance

Cloud Operations Disciplines



Inventory & Visibility

Establish a defined inventory of assets. Develop visibility into the asset telemetry.



Operational Compliance

Manage configuration drift and standards. Apply management automation and controls.



Protect & Recover

Implement solutions to minimize performance interruptions and ensure rapid recovery when needed.



Platform Operations

Customize operations to improve performance of the common platforms that support multiple workloads.



Workload Operations

Understand workload telemetry. Align workload operations to performance and reliability commitments.

Operations in Azure

- Azure Monitor
- Azure Automation / Serverless
- Azure Security Center
- Azure Service Health
- Azure Advisors

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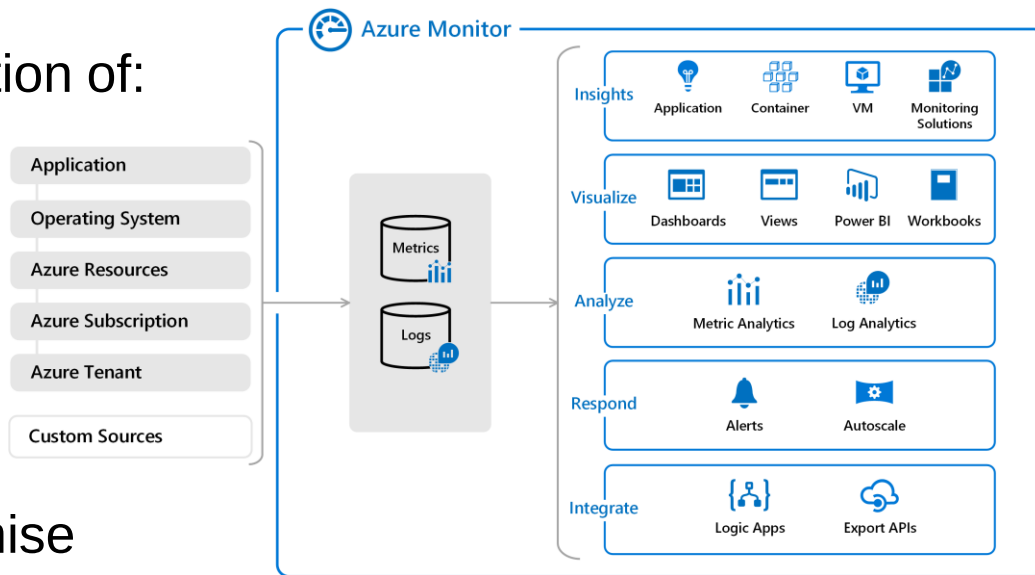
**Operations
Monitor**



Azure Monitor

- Functionality is a combination of:

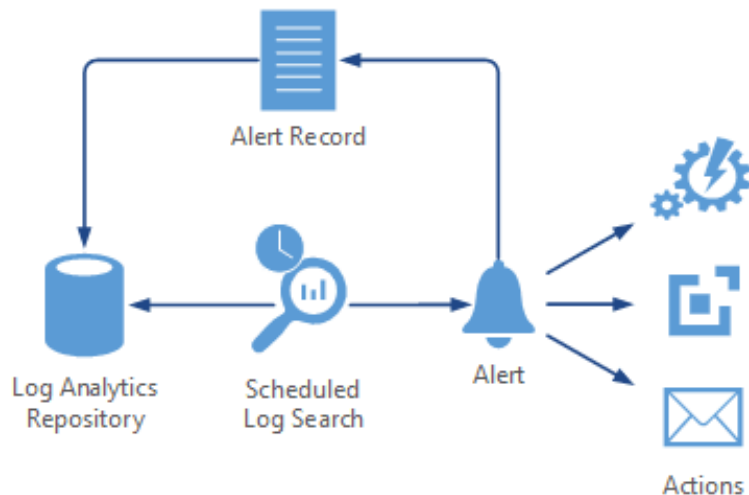
- » Log Analytics
- » Azure Metrics
- » Application Insights



- Can also monitor On-Premise servers (virtual machines

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Azure Alerts



- Triggered by Metrics, Activity log or KUSTO Query
- Can send an Email, API Call and a others
- Smart groups will aggregate alerts with Machine Learning

Azure Log Analytics

- Store and analyze data
- Agent available for most Operating Systems
- Uses KUSTO Language for Queries

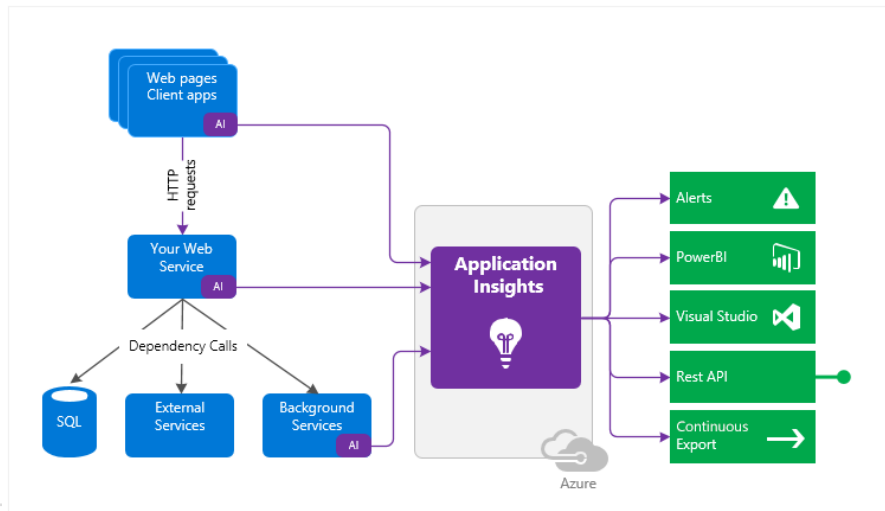
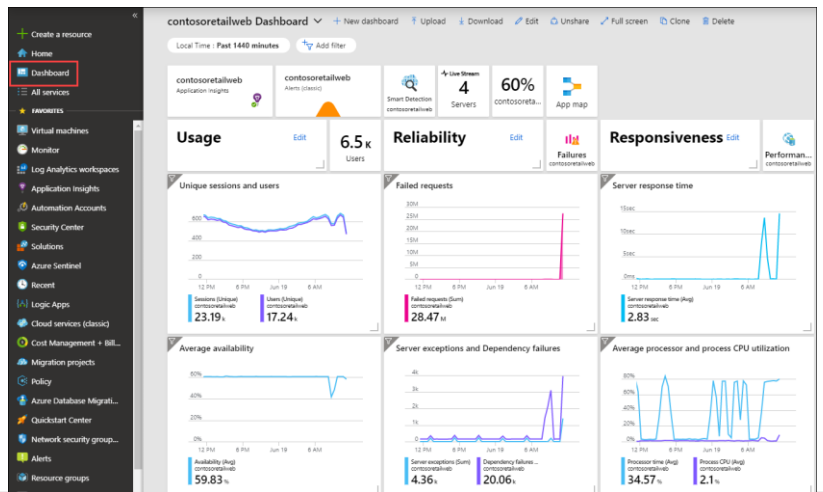
Kusto	 Copy
<pre>SecurityEvent where Level == 8 and EventID == 4672</pre>	

Integrate with existing monitoring system

- Log Analytics and Metrics can be consumed by 3rd party providers
 - » Splunk, DataDog, Grafana, ELK
- **Discussion**
- Do we need to integrate the Azure monitoring into existing systems?

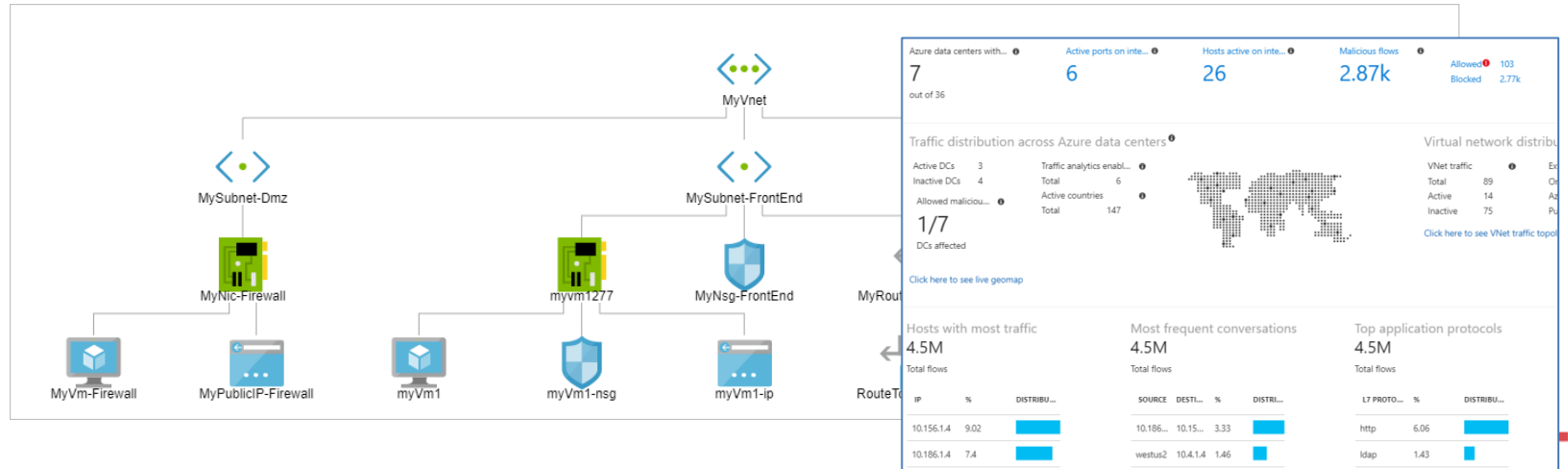
Application Insights

- Application Performance Monitoring tool
- Can be easily integrated within your Application or website



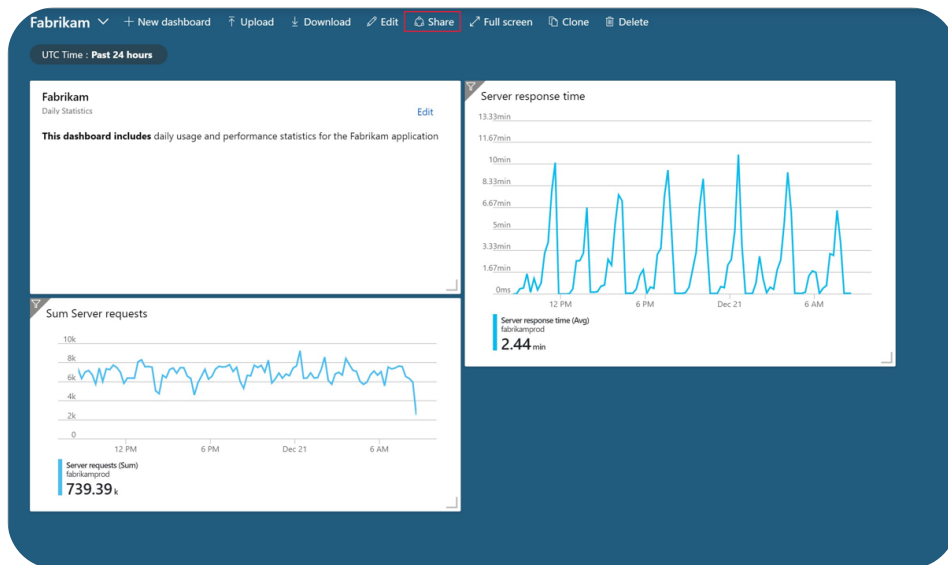
Network Watcher (IaaS)

- Monitor communication between VM and endpoints
- Diagnose issues with VM, NSG(firewall rules) and VPN
- Support for packet capture



Azure Dashboards

- Custom dashboards based on the information from Azure.



Guidance on Audit / Compliance logs

- Archive Azure AD Diagnostic logs for 365 days
 - » Sign-in
 - » Retention for 365 days
- Archive Azure Resource Manager Activity Logs
 - » Retention for 365 days

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Operations
Configure



Virtual Machine Updates

Schedule update deployment
 Add Azure VMs
 Add non-Azure machine
 Manage machines

Non-compliant machines

4
out of 6

Machines need attention (6)

Critical and security 4
Other 2
Not assessed 0

Missing updates (37)

Critical 1
Security 0
Others 36

Failed update deployments

2
out of 7 in the past six months

Learn more

[Update Management](#)

[Provide feedback](#)

Machines (6)
 Missing updates (37)
 Update deployments
 Scheduled update deployments

Filter by name

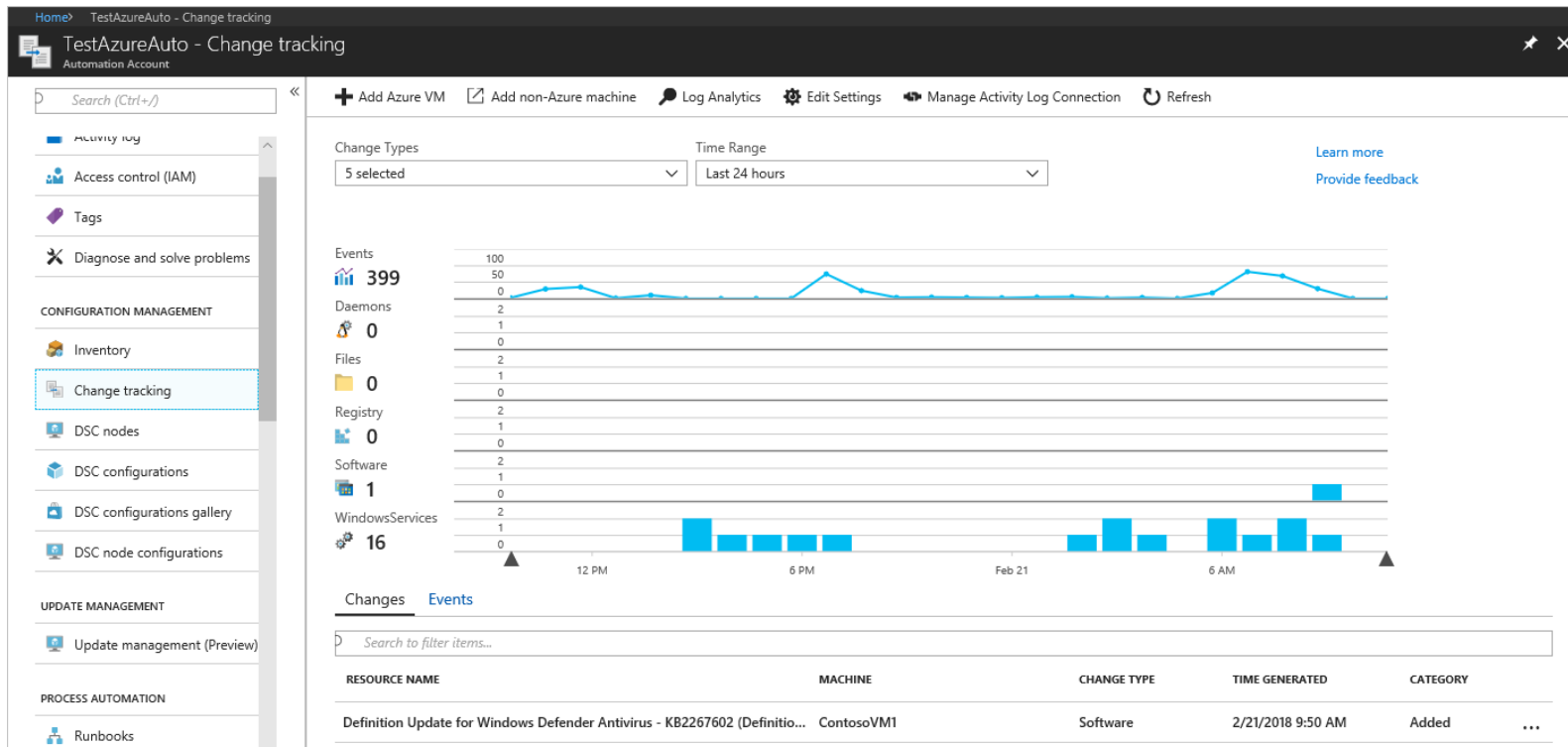
Compliance: All

Platform: All

Operating system: All

MACHINE NAME	COMPLIANCE	PLATFORM	OPERATING SYSTEM	CRITICAL MISSING UPDATES	SECURITY MISSING UPDATES	OTHER MISSING UPDATES	UPDATE AGENT READINESS
SQL01.internal.lab	Non-compliant as of 6/14/2018, 10:29 AM	Azure	Windows	1	0	5	Ready (view)
CAS01.internal.lab	Non-compliant as of 6/14/2018, 10:48 AM	Azure	Windows	1	0	4	Ready (view)
DC01.internal.lab	Non-compliant as of 6/14/2018, 5:23 AM	Azure	Windows	1	0	4	Ready (view)
WinVM6	Non-compliant as of 6/14/2018, 11:10 AM	Azure	Windows	1	0	3	Not configured (troubleshoot)
LinuxVM2	Compliant as of 6/14/2018, 11:09 AM	Azure	Linux	0	0	30	Ready (view)
LinuxVM6	Compliant as of 6/14/2018, 11:09 AM	Azure	Linux	0	0	30	Ready (view)

Change Tracking and Inventory



Azure Automation

- Run Python or Powershell script to perform tasks
 - » On Schedule
 - » On certain events
- Desired State Configuration

Serverless

- Azure functions / Logic apps
 - » Use language of choice
 - » Endless flexibility



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Operations
Security

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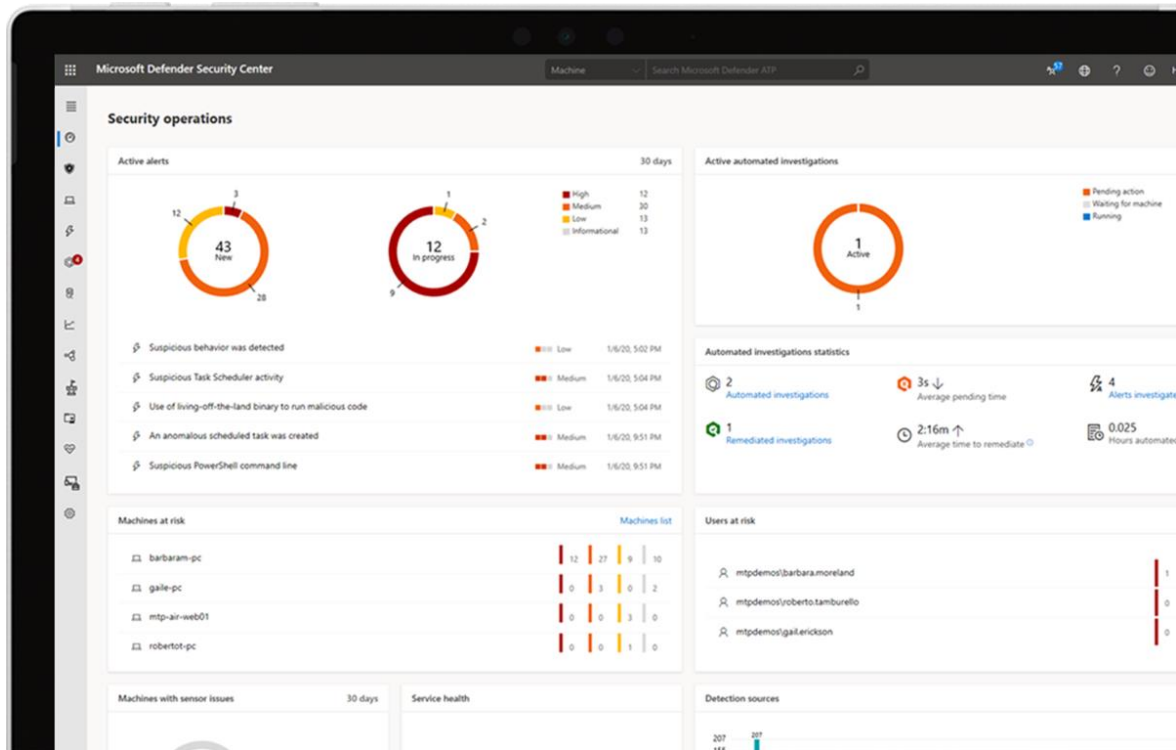
Security Center

■ Protect, Detect and Respond



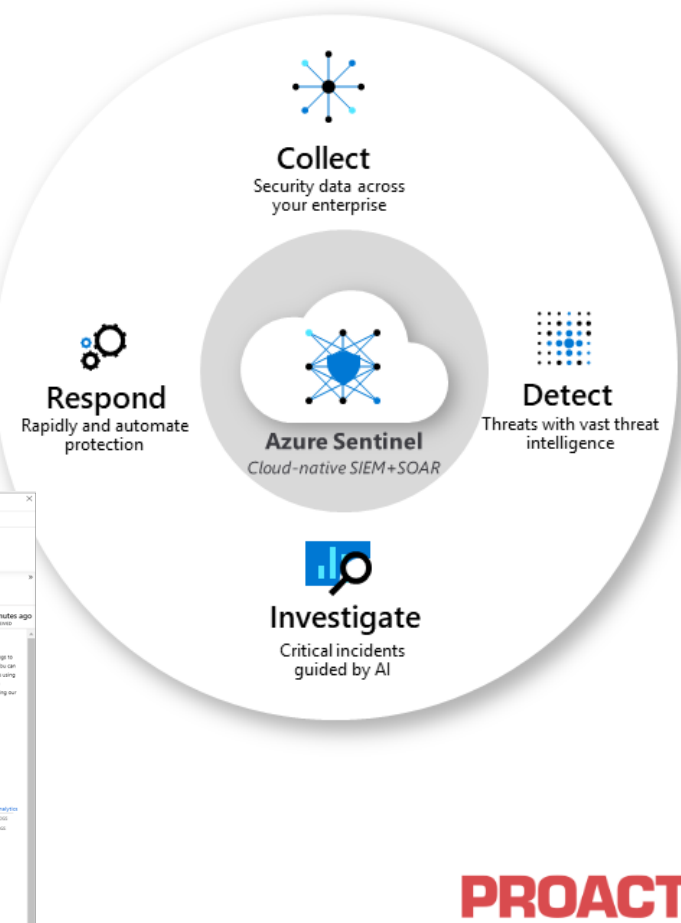
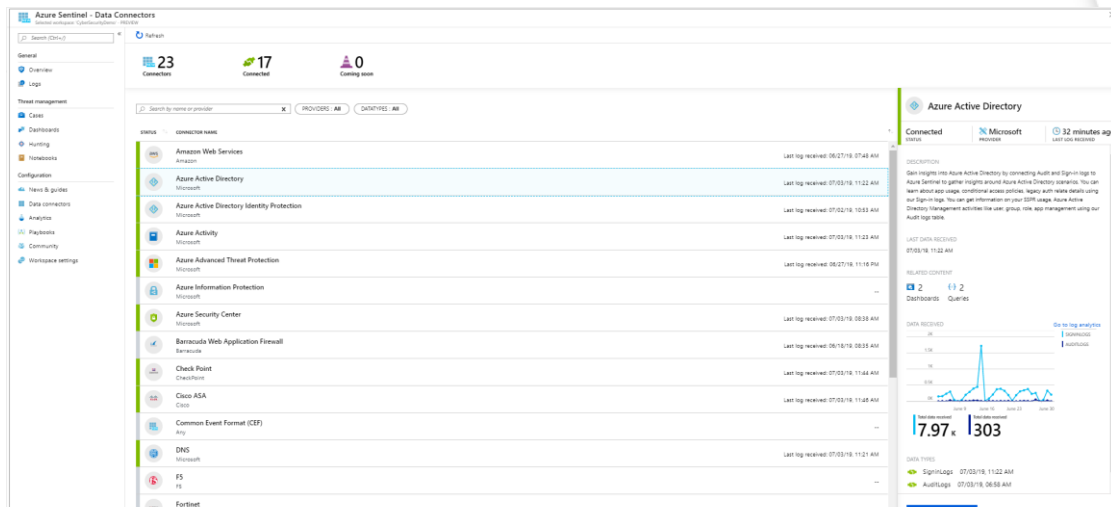
Advanced Threat Protection

- Threat & Vulnerability Management
- Attack Surface Reduction
- Next Generation Protection
- Endpoint Detection & Response
- Auto Investigation & Remediation
- Microsoft Threat Experts



Azure Sentinel

- security information event management (SIEM)
- security orchestration automated response (SOAR)



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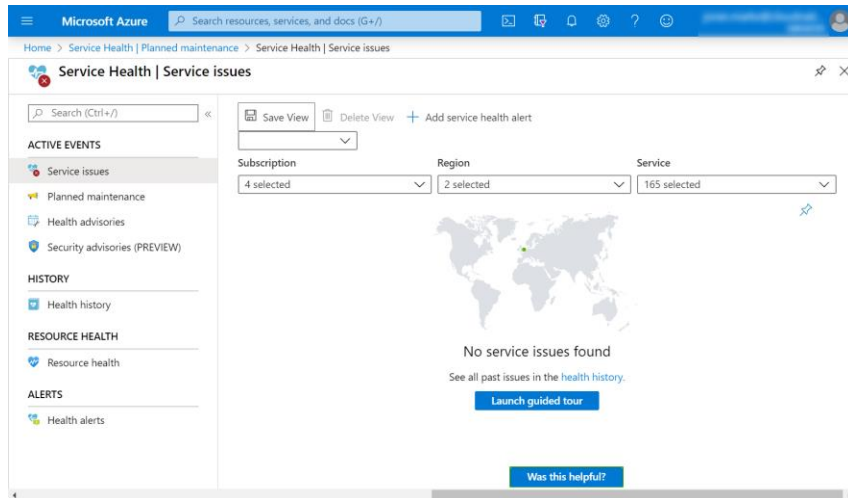
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Operations
Service Health



Platform Health

- Active
 - » Service issues
 - » Planned maintenance
 - » Health advisors
 - » Security advisor
- History
 - » Health history
- Alerts



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Operations Backup & DR



Recovery Service Vault

- Used for Backup & Site-Recovery
 - » Virtual machines
 - Can be used for On-Premise backup
-
- Cannot be used over different subscriptions!
 - Not possible to move this resource to another subscription

PaaS Service Backup

- Depends on the type of services
- Some services provide built-in capabilities
 - » Azure SQL Database
 - » Cosmos DB
 - » Azure Files

Disaster Recovery

- Should be part of the solution design
 - » Design for redundancy of service, zone or region failure
 - » Design for recoverability of service, zone or region failure
- Recovery Services can be used for VM's
 - » Onpremise
 - » Region
 - » Availability zone

Discussion

What kind of DR level are to be expected?

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Discussion



A low-angle, upward-looking photograph of several tall skyscrapers. The image is heavily stylized with a deep red color overlay. The word "PROACT" is written in large, bold, white, sans-serif capital letters across the center of the image. The perspective creates a sense of height and scale, with the buildings converging towards the top of the frame.

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